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CONTROL ARM FOR LAWN MOWER CONTROLS

Abstract

A control arm for a lawn mower, the control arm including a lower arm having a slot defined therein, an upper arm pivotably coupled to the lower arm, the upper arm having a cam surface defining a first recess and a second recess, and a latch mechanism formed between the upper and lower arms to selectively secure the upper arm in an operational position or a stored position. The latch mechanism includes a rod supported and movable within the slot, and a biasing member configured to bias the rod into engagement with the first recess to secure the upper arm in the operational position or the second recess to secure the upper arm in the stored position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application No. 63/337,818, filed on May 3, 2022, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present invention relates to a steering mechanism for lawn mower controls.

SUMMARY

[0003] In one aspect, the invention provides a control arm for a lawn mower. The control arm includes a lower arm having a slot defined therein, an upper arm pivotably coupled to the lower arm, the upper arm having a cam surface defining a first recess and a second recess, and a latch mechanism formed between the upper and lower arms to selectively secure the upper arm in an operational position or a stored position. The latch mechanism includes a rod supported and movable within the slot, and a biasing member configured to bias the rod into engagement with the first recess to secure the upper arm in the operational position or the second recess to secure the upper arm in the stored position.

[0004] In some aspects, the slot of the lower arm is a first slot, and the lower arm further includes a second slot that receives a pivot pin to pivotably couple the upper arm to the lower arm.

[0005] In some aspects, the lower arm includes a yoke having a first yoke arm, a second yoke arm, and a recess defined between the first and second yoke arms. A portion of the upper arm is received in the recess. In some aspects, the biasing member is a first biasing member, and the control arm further includes a second biasing member. A first sleeve is defined in the first yoke arm such that the first biasing member is received in the first sleeve of the first yoke arm. A second sleeve is defined in the second yoke arm such that the second biasing member is received in the second sleeve of the second yoke arm. In some aspects, the control arm further includes a journal bearing rotatably supported on the rod within the recess defined between the first and second yoke arms. The journal bearing is positioned between the rod and the cam surface. The journal bearing and the rod are configured to rotate relative to each other when the upper arm is moved between the operational and stored positions.

[0006] In some aspects, the upper arm further includes a first component, a second component, and an adjustable coupling that couples the first component with the second component. In some aspects, the first component includes a grip portion configured to be grasped by an operator to manipulate the control arm, and the adjustable coupling is operable to adjust the position of the grip portion relative to the operator.

[0007] In some aspects, the control arm further includes a brake cable coupled to the upper arm. The brake cable is operable to lock a ground engaging element of the lawn mower when the upper arm is in the stored position.

[0008] In some aspects, the control arm further includes a switch coupled to the lower arm, and the switch is in communication with a control unit on the lawn mower to selectively indicate to the control unit that the upper arm is in a stored position.

[0009] In another aspect, the invention provides a control arm for a lawn mower having ground engaging elements. The control arm includes a lower arm, an upper arm pivotably coupled to the lower arm, a latch mechanism formed between the upper and lower arms to selectively secure the upper arm in an operational position or a stored position, a sensor coupled to the lower arm, the sensor in communication with a control unit to selectively indicate to the control unit that the upper arm is in the stored position, an actuator coupled to the upper arm and configured to communicate with the sensor when the upper arm is in the stored position, and a brake actuation device coupled to the upper arm, the brake actuation device being operable to lock the ground engaging element when the upper arm is in the stored position.

[0010] In some aspects, the brake cable is coupled to the actuator, and the brake cable is configured to be actuated as the upper arm pivots toward the stored position.

[0011] In some aspects, the latch mechanism includes a rod supported and movable within a slot, and a biasing member configured to bias the rod into engagement with the first recess to secure the upper arm in the operational position or with the second recess to secure the upper arm in the stored position. In some aspects, the lower arm includes a yoke having a first yoke arm, a second yoke arm, and a recess defined between the first and second arms, and a portion of the upper arm is received in the yoke recess. In some aspects, the biasing member is a first biasing member, and the control arm further includes a second biasing member, a first sleeve defined in the first yoke arm, and a second sleeve defined in the second yoke arm. Each of the first biasing member and second biasing member is received in the first yoke arm and the second yoke arm.

[0012] In another aspect, the invention provides a method for assembling a control arm for a lawn mower. The method includes providing a lower arm having a sleeve and a slot that is defined orthogonal to the sleeve, inserting a biasing member into the sleeve, compressing the biasing member using a tool, inserting a rod into the slot while the biasing member is compressed, and pivotably coupling an upper arm to the lower arm after the rod has been inserted into the slot.

[0013] In some aspects, the sleeve is a first sleeve, the slot is a first slot, and the biasing member is a first biasing member, such that providing the lower arm includes providing a second sleeve and a second slot that is defined orthogonal to the second sleeve in the lower arm, and inserting the biasing member includes inserting the first biasing member into the first sleeve and inserting a second biasing member into the second sleeve.

[0014] In some aspects, compressing the biasing member within the sleeve using a tool includes inserting the tool into the slot over both the first biasing member and the second biasing member and pushing the tool against a bias of both the first biasing member and the second biasing member.

[0015] In some aspects, pivotably coupling the upper arm to the lower arm includes inserting a pivot pin through the second slot defined in the lower arm and through an aperture defined in the upper arm. In some aspects, pivotably coupling the upper arm to the lower arm includes providing the upper arm with a cam surface including a first recess, a second recess, and a protrusion extending between the first recess and the second recess, such that the method further includes releasing the compression of the biasing member such that the biasing member pushes the rod into engagement with one of the first recess and the second recess. In some aspects, inserting the rod into the slot while the biasing member is compressed includes providing a rod having a bearing mounted at a location on the rod that is configured to engage the one of the first recess and the second recess when the compression of the biasing member is released.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. **1** is a perspective view of a lawn mower including a latch mechanism and a multi Hirth joint according to the present invention.

[0017] FIG. **2** is a perspective view of two control arms of the lawn mower, each including the latch mechanism and the multi Hirth joint.

[0018] FIG. **3** is an exploded view of one of the control arms.

[0019] FIG. **4** is an exploded view of the multi Hirth joint.

[0020] FIG. **5** is a cross-sectional view of the latch mechanism taken through the center of the control arm.
[0021] FIG. **6**A is a cross-sectional view of the latch mechanism in a first position.
[0022] FIG. **6**B is a cross-sectional view of the latch mechanism in a second position.
[0023] FIG. **7** is a perspective view of the latch mechanism.
[0024] FIG. **8** is a perspective view of the control arm connected to a motor of the mower.
[0025] FIG. **9** is a flow diagram of a method for assembling the control arm.

DETAILED DESCRIPTION

[0026] Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways.

[0027] FIG. **1** illustrates a lawn mower **10** including a frame **15**, a prime mover **20**, an operator zone **25**, drive wheels **50**, casters **60**, and a mower deck assembly **70**. In the illustrated embodiment, the prime mover is an internal combustion engine supported by the frame **15** rearward the operator zone. In other embodiments, the prime mover **20** may be a bank of batteries in a rear storage compartment supported by the frame **15**. The operator zone **25** is supported by the frame **15** forward of the prime mover **20** and includes a seat **75** for the operator and all controls within reach of the operator while seated on the seat **75**. The operator zone **25** includes a left control arm **110′** and a right control arm **110″** which will be discussed in more detail below. The drive wheels **50** support a rear portion of the frame **15** and are rotated by drive motors **85** (FIG. **8**) under power from the prime mover **20** and at a speed and direction dictated by position of the control arms **110′**, **110″**. In the illustrated embodiment, the drive motors are hydraulic transaxles. The casters **60** are passive wheels that swivel about vertical axes. The mower deck assembly **70** is suspended from the frame **15** and includes a mower deck **80** having multiple blades under the mower deck **80**. In the illustrated embodiment, the blades are operably coupled to the prime mover **20**. In other embodiments, the mower deck assembly **70** includes one or more electric deck motors that drive rotation of the blades beneath the mower deck **80** under power from the prime mover **20**.

[0028] Other than a unique adjustable joint and the latch mechanism in the control arms **110′**, **110″**, the lawn mower **10** is of a type well known in the art and referred to as a zero turn radius mower. The terms left, right, forward, rearward, up and down and all other directional terms will be made from the point of view of an operator seated in the operator zone **25**. The lawn mower **10** includes a longitudinal axis “L” extending forward and rearward and a transverse axis “T” extending left and right perpendicular to the longitudinal axis L. A component is said to pivot left and right, sideways, outboard and inboard, or similar if it pivots about an axis that is parallel to the longitudinal axis L. A component is said to pivot forward and back, fore and aft, or similar if it pivots about an axis that is parallel to the transverse axis T. The term “outboard” indicates a direction parallel to the transverse axis T and away from the longitudinal axis L (i.e., left and right away from the mower **10**) and the term “inboard” indicates an opposite direction (i.e., toward the longitudinal axis L from one of the sides).

[0029] Turning to FIGS. **2** and **3**, the left and right control arms **110′**, **110″** are mirror images of each other and will be described together with reference to a control arm **110** (FIG. **3**) unless there is a need to distinguish them from each other. The control arm **110** includes a lower arm **115**, an upper arm **120**, and a latch mechanism **125** connecting the lower arm **115** and upper arm **120**. In the illustrated embodiment, the lower arm **115** pivots fore and aft and includes a linkage **130** to an associated hydraulic pump or transmission that operates an associated drive wheel **50**. In other embodiments, the lower arm **115** may include a position sensor that provides an input to a control unit **300** (FIG. **7**) to control the speed of the associated drive wheel **50** (e.g., drive by wire steering control). The upper end of the lower arm **115** includes a yoke **135** that defines a first yoke arm

135*a*, a second yoke arm **135***b*, and a recess **137** defined therebetween. The upper arm **120** includes a grip **140** that is grasped by the operator to manipulate the control arm **110**.

[0030] As illustrated in FIG. **5**, the lower arm **115** further includes a first elongated slot **145**, a second elongated slot **150**, a first blind bore or sleeve **155**, and a second blind bore or sleeve **160**. The first slot **145** is positioned between the frame **15** (FIG. **1**) and the upper end of the lower arm **115**. The second slot **150** is positioned between the first slot **145** and the upper end of the lower arm **115**. The first slot **145** and the second slot **150** each extend through the yoke arms **135***a*, **135***b* of the lower arm **115** in a direction parallel to the longitudinal axis L (FIG. **1**).

[0031] The first sleeve **155** is defined at least partially in a first yoke arm **135***a*. The second sleeve **160** is defined at least partially in a second yoke arm **135***b*. Each of the first sleeve **155** and the second sleeve **160** extends (i.e., has a longitudinal axis extending) in a direction orthogonal (e.g., perpendicular) to both the transverse axis T (FIG. **1**) and the longitudinal axis L (FIG. **1**). Therefore, the first sleeve **155** extends through, or intersects, the first slot **145** and the second slot **150** in the first yoke arm **135***a*. The second sleeve **160** extends through, or intersects, the first slot **145** and the second slot **150** in the second yoke arm **135***b*. Further, the first and second sleeves **155**, **160** are equally spaced on opposing sides of the recess **137** (e.g., the center of the lower arm **115**) defined by the yoke **135**.

[0032] Referring to FIGS. **3** and **4**, the upper arm **120** further includes a first component **165**, a second component **170**, and an adjustable coupling mechanism **175** connecting the first component **165** and the second component **170**. The first component **165** includes the grip **140** that is grasped by the operator to manipulate the control arm **110**. The second component **170** includes an adjustment slot **180** positioned at a first end **170***a* of the second component **170** and a cam surface **185** positioned at a second end **170***b* of the second component **170**. The adjustment slot **180** receives the adjustable coupling mechanism **175** to couple the first component **165** to the second component **170**. The adjustable coupling mechanism **175** may be adjusted along a length of the adjustment slot **180** to adjust a height of the first component **165** relative to the second component **170**. As such, the adjustment slot **180** enables adjustment of the height of the grip **140** relative to the seat **75** (FIG. **1**) to accommodate operators of different heights. As best illustrated in FIGS. **6**A and **6**B, the cam surface **185** includes a first recess **190**, a second recess **195**, and a protrusion **197** defined between the first and second recesses **190**, **195** and at least partially separating the first recess **190** and the second recess **195**.

[0033] Returning reference to FIGS. **3** and **4**, the adjustable coupling mechanism **175** includes a multiple Hirth interface **200** positioned on the second component **170** and a single Hirth interface **205** positioned on the first component **165**. The multiple Hirth interface **200** includes a lower Hirth interface **200***a*, a middle Hirth interface **200***b*, and an upper Hirth interface **200***c*. Each of the lower, middle, and upper Hirth interfaces **200***a*, **200***b*, **200***c* includes face teeth **210** arranged in a partial circular pattern in a face or side surface of the multiple Hirth interface **200**. The adjustment slot **180** is arranged on the multiple Hirth interface **200**. Specifically, the adjustment slot **180** extends diametrically through the middle Hirth interface **200***b* and has opposite ends centered in the lower Hirth interface **200***a* and upper Hirth interface **200***c*.

[0034] The single Hirth interface **205** has a circular pattern of face teeth **215** arranged in a circular pattern in a face or side surface of the circular Hirth interface **205**. The face teeth **215** are sized and shaped to mesh or mate with the face teeth **210** of the lower, middle, and upper Hirth interfaces **200***a*, **200***b*, **200***c*. The single Hirth interface **205** includes a through-hole **220** centered in the circular pattern of face teeth **215**. When properly assembled, the single Hirth interface **205** confronts the multiple Hirth interface **200** so the respective face teeth **210**, **215** can mate in multiple discrete angular positions about the transverse axis T. As such, the single Hirth interface **205** and the multiple Hirth interface **200** further enable angular adjustment of the grip **140** relative to the seat **75** (FIG. **1**) to accommodate operators with different arm lengths. The adjustable coupling mechanism **175** further includes a knob **225** and a coupling rod **230**. The coupling rod **230** may be inserted through the adjustment slot **180** and the through-hole **220** to set a position of the first component **165** relative to the second component **170**. The knob **225** may be tightened to secure the position of the first component **165** relative to the second component **170**.

[0035] Now with reference to FIG. **5**, the latch mechanism **125** includes a pivot pin **235**, a rod **240**, a first biasing member **245**, a second biasing member **250**, and a sleeve or journal bearing **263** rotatably supported on the rod **240**. The second component **170** of the upper arm **120** is inserted within the recess **137** defined between the yoke arms **135***a*, **135***b* so that the cam surface **185** is disposed within the recess **137** facing the bottom of the recess **137**. The second component **170** is coupled to the lower arm **115** by the pivot pin **235**. In the illustrated embodiment, the pivot pin **235** is received in the second slot **150** of the lower arm **115** and an aperture **252** (as best illustrated in FIG. **3**) that extends through the upper arm **120** to pivotally couple the upper arm **120** to the lower arm **115**.

[0036] The rod **240** is received in the first elongated slot **145** of the lower arm **115**. The first biasing member **245** is received in the first sleeve **155** of the lower arm **115**. The second biasing member **250** is received in the second sleeve **160** of the lower arm **115**. As such, the first biasing member **245** and the second biasing member **250** may engage the rod **240** to bias the rod **240** toward the upper end of the lower arm **115** and into engagement with the cam surface **185** (as best illustrated in FIG. **3**) defined in the upper arm **120**. The journal bearing **263** is rotatably supported on the rod **240** in the recess **137** defined between the yoke arms **135***a*, **135***b*. The journal bearing **263** and the rod **240** may rotate relative to each other as the journal bearing **263** engages the cam surface **185**, which allows for a smoother rolling action and reduces wear on the cam surface **185** as the upper arm **120** moves relative to the lower arm **115**.

[0037] As illustrated in FIG. **7**, the control arm **110** further includes a sensor **255** coupled to the lower arm **115**, an actuator **260** coupled to the second component **170** of the upper arm **120** (FIG. **2**), and a brake actuation device **265** coupled to the second component **170** of the upper arm **120** (FIG. **2**). In the illustrated embodiment, the sensor **255** is a contact switch having a plunger **257** that is selectively actuated by the actuator **260** when the upper arm **120** is moved to a stored position (e.g., illustrated by the left control arm **110**′ in FIG. **2**). The sensor **255** is electrically connected (e.g., by a connector **259**) to a control unit **300** of the mower **10** (FIG. **1**). The actuator **260** is rotatable with the upper arm **120** and is configured to actuate or communicate with the sensor **255**. In the illustrated embodiment, the actuator **260** is an elongated plate coupled to the upper arm **120**. In other embodiments, the actuator **260** may be any other structure configured to rotate with the second component **170** and communicate the sensor **255**. For example, the actuator **260** may be integrally formed on the upper arm **120**. In other embodiments, the sensor **255** may be a non-contact switch such as a magnetic switch, an optical sensor, or the like. In such embodiments, the actuator **260** may communicate with the non-contact switch to communicate to the control unit **300** that the upper arm **120** is in the stored position.

[0038] The brake actuation device **265** is operable to lock the drive wheels **50** (FIG. **1**). In the illustrated embodiment, the brake actuation device **265** is actuated by rotation of the second component **170** to lock the drive wheels **50** (FIG. **1**). In other embodiments, the brake actuation device **265** may be an electronic brake mechanism that is actuated in response to the upper arm **120** being positioned in the stored position. The brake actuation device **265** may be coupled directly to the actuator **260** such that the brake actuation device **265** and the sensor **255** are simultaneously actuated with rotation of the actuator **260**.

[0039] Returning reference to FIGS. **5** and **6**A, an operator may position the control arm **110** in an operational position (the right control arm **110**″ in FIG. **2**) or a stored position (the left control arm **110**′ in FIG. **2**). In the operational position, the second component **170** of the upper arm **120** extends substantially parallel with the lower arm **115**. Further, each of the first biasing member **245** and the second biasing member **250** biases the rod **240** into engagement with the first recess **190** to secure the upper arm **120** in the operational position. Turning reference to FIG. **3**, when the control arm **110** is in the operational position, a user may grasp the grip **140** to manipulate the control arm **110**. An operator may move the upper arm **120** forward to pivot the lower arm **115** in the fore direction thereby driving the drive wheels **50** (FIG. **1**) forward. Alternatively, an operator may move the upper arm **120** rearward to pivot the lower arm **115** in the aft direction thereby driving the drive wheels **50** (FIG. **1**) rearward.

[0040] As illustrated in FIGS. **5**, **6**B, and **7**, to switch the position of the control arm **110** from the operational position to the stored position, an operator may pivot the upper arm **120** of the control arm outboard (e.g., control arm **110′** in FIG. **2**). Specifically, as the upper arm **120** is pivoted outboard, the rod **240** compresses the first biasing member **245** and the second biasing member **250**. Compression of the biasing members **245**, **250** causes the rod **240** to move across the protrusion **197** and into engagement with the second recess **195**, which secures the upper arm **120** in the stored position. The protrusion **197** may provide some resistance for the rod **240** so that the rod **240** is inhibited from moving between the recesses **190**, **195** unless the control arm **110** is operator-adjusted from the operational position to the stored position. The journal bearing **263** and the rod **240** may rotate relative to each other as the upper arm **120** is moved to the stored position. Movement of the upper arm **120** to the stored position also causes the actuator **260** (FIG. **7**) to actuate the sensor **255** and the brake actuation device **265**. Specifically, the actuator **260** depresses the sensor **255** as the control arm **110** is pivoted from the operational position to the stored position. By actuating the sensor **255**, the sensor **255** indicates to the control unit **300** of the mower **10** (FIG. **1**) that the mower **10** (FIG. **1**) is in the parked position. By actuating the brake actuation device **265**, the brake actuation device **265** engages a pawl **270** (FIG. **8**) on the drive motor **85** to lock the drive wheels **50** (FIG. **1**). The indication provided by the sensor **255** and the locked drive wheels **50** (FIG. **1**) allows the prime mover **20** to remain running even when in the parked position. In absence of the indication provided by the sensor **255** and the locked drive wheels **50** (FIG. **1**), a typical mower would shut off the prime mover **20** to inhibit unintended use of the mower **10**. By locking the drive wheels **50** (FIG. **1**) and indicating that the mower **10** is in a parked position, the drive wheels **50** (FIG. **1**) are prevented from moving and mower blade rotation is stopped. This feature allows an operator to pause and resume operation of the mower **10** (FIG. **1**) without having to restart the prime mover **20**. As such, this feature may reduce overall fuel consumption of the mower **10** (FIG. **1**).

[0041] FIG. **9** illustrates an assembling process **500**. The assembling process **500** is a method for assembling the steering mechanism. Although the assembling process **500** is described with reference to these steps, not all of the steps need to be performed or need to be performed in the order presented.

[0042] The assembling process **500** includes, with reference to FIGS. **5** and **9**, providing the lower arm **115** having the first sleeve **155**, the second sleeve **160**, the first slot **145**, and the second slot **150** (Step **510**). The first slot **145** and the second slot **150** are defined orthogonal to the first sleeve **155** and the second sleeve **160**.

[0043] The first biasing member **245** is inserted into the first sleeve **155** and the second biasing member **250** into the second sleeve **160** (Step **520**). Each of the first biasing member **245** and the second biasing member **250** may be inserted such that an end of the first biasing member **245** and the second biasing member **250** extends from the first sleeve **155** and the second sleeve **160**, respectively, into the first elongated slot **145**. In such embodiments in which only one of the first sleeve **155** and the second sleeve **160** is provided, the lower arm **115** may be provided with just one of the first biasing member **245** and the second biasing member **250** that corresponds to the one of the first sleeve **155** and the second sleeve **160**.

[0044] With continued reference to FIGS. **5** and **9**, the first biasing member **245** and the second biasing member **250** are each compressed with a tool (Step **530**). In the illustrated embodiment, the biasing members **245**, **250** are each compressed by inserting a tool through the first slot **145**. For example, the tool may be a screwdriver, a flathead screwdriver, or any other similar relatively small diameter tool. The tool may be inserted into the first elongated slot **145** over a top of the first biasing member **245** and the second biasing member **250**. The tool may then be pushed in a direction away from the upper end of the lower arm **115** to compress the first biasing member **245** and the second biasing member **250**. For example, the first biasing member **245** and the second biasing member **250** may be compressed such that the end of the first biasing member **245** and the second biasing member **250** does not extend into the first elongated slot **145**.

[0045] The rod **240** is then inserted into the first elongated slot **145** while each of the first biasing member **245** and the second biasing member **250** is compressed (Step **540**). Compressing the first biasing member **245** and the second biasing member **250** provides clearance in the first elongated slot **145** for the rod **240** to be fully inserted into the first elongated slot **145**. In the illustrated embodiment, the journal bearing **263** is inserted in the recess between the yoke arms **135***a*, **135***b* and the rod **240** is inserted through the journal bearing **263** as the rod **240** is inserted into the first elongated slot **145**.

[0046] The upper arm **120** is pivotably coupled to the lower arm **115** (Step **550**). Specifically, the aperture **252** in the upper arm **120** may be aligned with the second slot **150** defined in the yoke arms **135***a*, **135***b*. The pivot pin **235** may then be inserted through the second slot **150** and the aperture **252** defined in the upper arm **120** to pivotably couple the upper arm **120** to the lower arm **115**. Once the upper arm **120** is coupled to the lower arm **115**, the first biasing member **245** and the second biasing member **250** biases the rod **240** toward the upper end of the lower arm **115** and into engagement with the cam surface **185** defined in the upper arm **120**

[0047] The foregoing is a description of one embodiment of the invention and is not limiting. The invention may be embodied in any outdoor power equipment or other application in which it is desired to provide linear and angular adjustment of one component with respect to another component in a single joint. Also, as one of ordinary skill in the art will appreciate, the components of the disclosed embodiment can be reversed and interchanged.

[0048] Thus, the invention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention as defined by the following claims.

Claims

1. A control arm for a lawn mower, the control arm comprising: a lower arm having a slot defined therein; an upper arm pivotably coupled to the lower arm, the upper arm having a cam surface defining a first recess and a second recess; and a latch mechanism formed between the upper and lower arms to selectively secure the upper arm in an operational position or a stored position, the latch mechanism including a rod supported and movable within the slot, and a biasing member configured to bias the rod into engagement with the first recess to secure the upper arm in the operational position or the second recess to secure the upper arm in the stored position.

2. The control arm of claim 1, wherein the slot of the lower arm is a first slot, the lower arm further including a second slot that receives a pivot pin to pivotably couple the upper arm to the lower arm.

3. The control arm of claim 1, wherein the lower arm includes a yoke having a first yoke arm, a second yoke arm, and a recess defined between the first and second yoke arms, and wherein a portion of the upper arm is received in the recess between the first and second yoke arms.

4. The control arm of claim 3, wherein the biasing member is a first biasing member, the control arm further including a second biasing member, wherein a first sleeve is defined in the first yoke arm such that the first biasing member is received in the first sleeve of the first yoke arm, and wherein a second sleeve is defined in the second yoke arm such that the second biasing member is received in the second sleeve of the second yoke arm.

5. The control arm of claim 3, further comprising a journal bearing rotatably supported on the rod within the recess defined between the first and second yoke arms, wherein the journal bearing is positioned between the rod and the cam surface, and wherein the journal bearing and the rod are configured to rotate relative to each other when the upper arm is moved between the operational and stored positions.

6. The control arm of claim 1, wherein the upper arm further includes a first component, a second component, and an adjustable coupling that couples the first component with the second component.

7. The control arm of claim 6, wherein the first component includes a grip portion configured to be grasped by an operator to manipulate the control arm, and wherein the adjustable coupling is operable to adjust the position of the grip portion relative to the operator.

8. The control arm of claim 1, further comprising a brake cable coupled to the upper arm, and wherein the brake cable is operable to lock a ground engaging element of the lawn mower when the upper arm is in the stored position.

9. The control arm of claim 1, further comprising a switch coupled to the lower arm, and wherein the switch in communication with a control unit on the lawn mower to selectively indicate to the control unit that the upper arm is in the stored position.

10. A control arm for a lawn mower having ground engaging elements, the control arm comprising: a lower arm; an upper arm pivotably coupled to the lower arm; a latch mechanism formed between the upper and lower arms to selectively secure the upper arm in an operational position or a stored position; a sensor coupled to the lower arm, the sensor in communication with a control unit on the lawn mower to selectively indicate to the control unit that the upper arm is in the stored position; an actuator coupled to the upper arm and configured to communicate with the sensor when the upper arm is in the stored position; and a brake actuation device coupled to the upper arm, the brake actuation device being operable to lock the ground engaging element when the upper arm is in the stored position.

11. The control arm of claim 10, wherein the brake cable is coupled to the actuator, and wherein the brake cable is configured to be actuated as the upper arm pivots toward the stored position.

12. The control arm of claim 10, wherein the latch mechanism includes a rod supported and movable within a slot, and a biasing member configured to bias the rod into engagement with the first recess to secure the upper arm in the operational position or with the second recess to secure the upper arm in the stored position.

13. The control arm of claim 12, wherein the lower arm includes a yoke having a first yoke arm, a second yoke arm, and a yoke recess defined between the first and second arms, and wherein a portion of the upper arm is received in the yoke recess.

14. The control arm of claim 13, wherein the biasing member is a first biasing member, the control arm further including a second biasing member, a first sleeve defined in the first yoke arm, and a second sleeve defined in the second yoke arm, each of the first biasing member and the second biasing member being received in the first yoke arm and the second yoke arm.

15. A method for assembling a control arm for a lawn mower, the method comprising: providing a lower arm having a sleeve and a slot that is defined orthogonal to the sleeve; inserting a biasing member into the sleeve; compressing the biasing member within the sleeve using a tool; inserting a rod into the slot while the biasing member is compressed; and pivotably coupling an upper arm to the lower arm after the rod has been inserted into the slot.

16. The method of claim 15, wherein the sleeve is a first sleeve, the slot is a first slot, and the biasing member is a first biasing member, wherein providing the lower arm includes providing a second sleeve and a second slot that is defined orthogonal to the second sleeve in the lower arm, and wherein inserting the biasing member includes inserting the first biasing member into the first sleeve and inserting a second biasing member into the second sleeve.

17. The method of claim 16, wherein compressing the biasing member within the sleeve using a tool includes inserting the tool into the slot over both the first biasing member and the second biasing member and pushing the tool against a bias of both the first biasing member and the second biasing member.

18. The method of claim 16, wherein pivotably coupling the upper arm to the lower arm includes inserting a pivot pin through the second slot defined in the lower arm and through an aperture defined in the upper arm.

19. The method of claim 18, wherein pivotably coupling the upper arm to the lower arm includes providing the upper arm with a cam surface including a first recess, a second recess, and a protrusion extending between the first recess and the second recess, and wherein the method further includes releasing the compression of the biasing member such that the biasing member pushes the rod into engagement with one of the first recess and the second recess.

20. The method of claim 19, wherein inserting the rod into the slot while the biasing member is compressed includes providing a bearing mounted at a location on the rod that is configured to engage the one of the first recess and the second recess when the compression of the biasing member is released.
